

Beyond Pair Spacing: Loss Scale and Startup Updates in Consistency Tuning

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12 September 2026

Abstract

Changing the training gap in a consistency model can alter both target construction and loss scale, making a quality improvement difficult to attribute. We separate these operations in Easy Consistency Tuning and follow their effects from local updates to generation quality. In the studied CIFAR-10 $q = 256$ configuration, an early denominator-only history improves quality after a common continuation and evaluation-EMA rebuilding, without enlarging the target-time gap. This observation motivates startup learning-rate interventions based on RAdam’s known scale response. In a responder-enriched eight-seed cohort, a first-five-update reduction improves endpoints, and moving the same nominal reduction to the next five updates produces worse outcomes. A fresh eight-seed $q = 128$ cohort, however, does not reproduce the predicted startup benefit or provide stable support for the denominator-history advantage. Existing telemetry confirms that the prescribed interventions were executed and changed local parameter movement; it does not explain the difference in endpoint response. The evidence therefore identifies a persistent scale-history effect and a window-dependent operation in specific $q = 256$ cohorts, while limiting their transfer to a new spacing setting and cohort. Local scaling identities specify testable interventions, but neither those identities nor their successful execution establish a general quality mechanism.

1 Introduction

Consistency models (CMs) offer generation in one or a few network evaluations by learning to map noisy states directly to a common endpoint (Song et al., 2023). Their fast inference depends on a demanding training problem: predictions at neighboring noise levels must agree without collapsing to an uninformative solution. Easy Consistency Tuning (ECT) makes this problem more accessible by initializing from a pretrained diffusion model and progressively tightening consistency constraints (Geng et al., 2025). Subsequent work improves training through variance reduction, time-range truncation, and better loss geometry (Wang et al., 2024; Lee et al., 2024; Kim et al., 2025). Understanding what actually changes when a training control is adjusted is therefore relevant to both CM methodology and its implementations.

We examine the training pair spacing. In an inverse-gap objective, enlarging this spacing changes two computations: the lower-noise input used to construct the self-generated target, and the denominator that scales the loss. ECT already recognizes the relationship between mapping and weighting and studies alternative weights. Our question concerns the origin of a measured quality benefit: does it require changing the target-time rule, or can a loss-scale operation alone produce it? We further ask whether a specific change in the optimizer’s actual updates has a corresponding endpoint consequence.

The distinction is sharper than a generic claim that weighting matters. Reweighting different noise levels can change the direction of the expected training gradient. In the configuration studied here, a denominator-only multiplier instead scales every same-state gradient by the same positive constant. It preserves its direction and the target-time rule. Adaptive normalization might appear to remove such scaling, yet the implemented optimizer also has

a startup branch, accumulated moments, and finite-precision execution. Their effects must be examined at the level of actual updates and resulting trajectories.

Our experiments proceed from component attribution to a targeted update test. An early joint spacing intervention yields an 8.60% endpoint FID improvement across 26 complete pairs after both histories continue under the same rule. A four-arm extension separates target construction from loss scaling: the denominator-only history yields an 8.75% improvement across 14 complete pairs, even when the evaluation EMA is rebuilt at the boundary. These results motivate learning-rate interventions specified from the startup response of RAdam. In the completed eight-seed pilot, lowering only A’s first five successful-update learning rates improves every seed; reversing the scale reduction in D worsens six of eight seeds. The two average directions match the hypothesis, with stronger evidence on the A side. An adjacent-window extension then asks whether the same reduction has the same consequence when delayed by five successful updates. It does not: the early window has lower endpoint FID in all eight paired comparisons, with delayed/early geometric FID ratio 1.11248. The evidence therefore concerns an operation whose placement matters, beyond the observation that an early schedule leaves a lasting effect.

The fresh $q = 128$ experiment adds an important limit to this account. All four paths complete for eight response-unselected seeds, yet neither the primary startup contrast nor the supportive denominator-history contrast identifies a stable quality advantage. The implemented LR windows and local update responses are present. Thus the new result limits the scope of the earlier evidence without establishing why the endpoint responses differ: both the spacing parameter and the cohort have changed.

We contribute (i) executable target and scale controls derived from a same-state gradient identity; (ii) evidence that a denominator-only history can improve common-continuation quality in the studied $q = 256$ setting; and (iii) startup and adjacent-window interventions that trace a possible route to this effect, together with a fresh cohort that limits its transfer. The theory supplies local relations and experimental motivation. The experiments establish operation-level effects and their observed limits, rather than a complete mediation account.

2 Background: consistency tuning and its controls

2.1 From diffusion sampling to a consistency function

A diffusion model learns to reverse a noising process, usually requiring multiple network evaluations to generate a sample. Its associated probability-flow ODE has the same time marginals as the diffusion SDE (Song et al., 2021). An ideal consistency function maps every point on one such ODE trajectory to the same data-end state, anchored by a boundary condition such as $f(x, 0) = x$. Evaluating that function on a high-noise input then supports one-step generation. Consistency distillation uses a pretrained diffusion model to construct training targets along approximate ODE steps; consistency training can instead construct pairs from data and shared noise (Song et al., 2023; Song & Dhariwal, 2024).

For the variance-exploding parameterization used here, the practical pair construction is

$$x_t = x_0 + t\epsilon, \quad x_r = x_0 + r\epsilon, \quad 0 \leq r < t. \quad (1)$$

The higher-noise prediction is trained toward a lower-noise prediction whose gradient is stopped. In our configuration, both branches use the current network parameters. This self-generated target is distinct from the EMA used to evaluate sample quality. Shared-noise pairs are a training construction; a particular line $x_0 + t\epsilon$ is not generally an exact probability-flow ODE trajectory.

2.2 What ECT controls

ECT starts from a pretrained diffusion model and uses a mapping $r = r(t)$ to choose the paired noise level. Its broader curriculum progressively tightens the consistency constraint. We isolate a fixed stage of this implementation: the baseline unclipped relative gap is

$$\frac{t - r_1}{t} = \frac{n(t)}{q}, \quad n(t) = 1 + \frac{8}{1 + \exp(t)}. \quad (2)$$

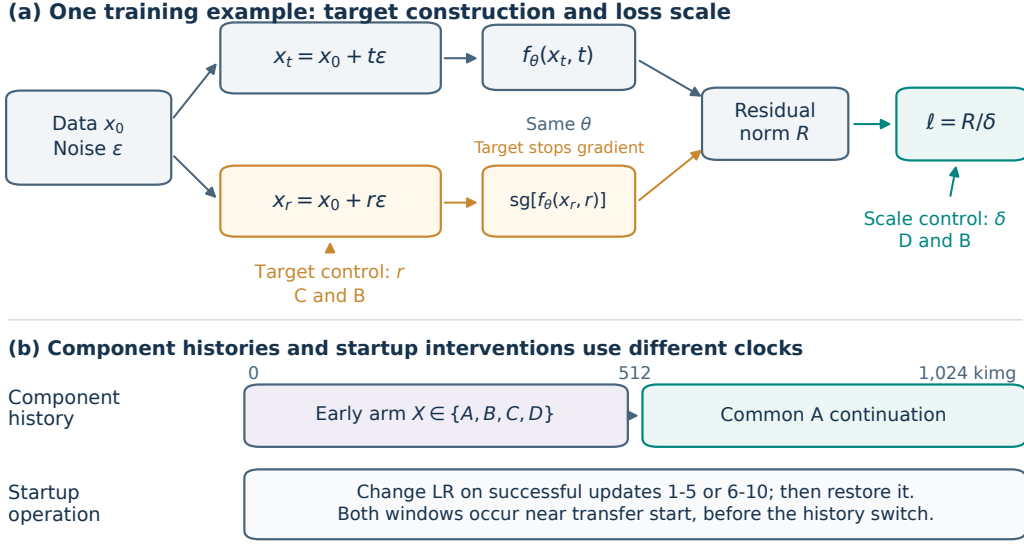


Figure 1: The training computation has two separately executable controls. The two noisy inputs share x_0 and ϵ . Their predictions use the same network parameters, with the lower branch stopped during backpropagation. Changing r changes the target input; changing δ scales the residual loss. A/B/C/D cross these controls. The bottom panel distinguishes an early component history from the five-successful-update LR interventions and their two placements. Neither changes the sampling NFE.

Here q is a training-spacing parameter (256 in the original studies and 128 in the fresh cohort); it is not the number of sampling steps. Increasing q makes pairs closer at a fixed noise level and stage. The joint intervention uses $t - r_g = g(t - r_1)$ with $g = 1.1$. This stage-specific formula does not describe every stage of ECT’s general curriculum.

A general consistency objective combines a distance between predictions with a weight. The main configuration uses inverse-gap weighting and the $c = 0$ limit of its norm-based distance:

$$\ell(r, \delta; \theta) = \frac{\|e_r\|_2}{\delta}, \quad e_r = f_\theta(x_t, t) - \underbrace{\text{sg}[f_\theta(x_r, r)]}_{\text{target output}}, \quad \delta > 0. \quad (3)$$

Normally $\delta = t - r$. We control r and δ separately: the former sets the target input, while the latter sets a multiplicative loss scale (Figure 1). The result is specific to this objective; ECT also studies weights that do not depend on the gap (Geng et al., 2025).

Three different clocks. Noise levels t, r index the inputs within an example. Training progress indexes attempts or successful optimizer updates; AMP can skip an attempted update. Sampling NFE counts network evaluations when generating an image. Our “first five updates” refer to successful updates at the beginning of ECT transfer tuning; the delayed control uses successful updates 6–10. Both occur near transfer start, whereas the history switch occurs at 512 king. One king denotes one thousand training images; attempted images count toward the budget even when AMP skips the update.

3 Separating target construction from loss scale

3.1 A local identity defines the interventions

Lemma 1 (Finite-gap, same-state gradient decomposition). Fix the parameters, student input, and random realization. Assume a differentiable student and finite nonzero residuals.

Table 1: Four early operations. The baseline gap is $\delta_1 = t - r_1$; $\delta_g = g\delta_1$ with $g = 1.1$ in real arithmetic on unclipped pairs.

Arm	Target input gap	Loss denominator	Changed computation
A	δ_1	δ_1	Baseline
B	$g\delta_1$	$g\delta_1$	Target and scale
C	$g\delta_1$	δ_1	Target only
D	δ_1	$g\delta_1$	Scale only

Write $J_t = \partial f_\theta(x_t, t)/\partial\theta$, $u_r = e_r/\|e_r\|_2$, and $s = \delta_1/\delta_g$. The implemented single-sided gradient satisfies

$$G(r, \delta) = \delta^{-1} J_t^\top u_r, \quad (4)$$

$$G(r_g, \delta_g) = sG(r_1, \delta_1) + \underbrace{\delta_g^{-1} J_t^\top (u_{r_g} - u_{r_1})}_{\text{target-dependent correction}}. \quad (5)$$

The identity follows by adding and subtracting $\delta_g^{-1} J_t^\top u_{r_1}$. Its role is to define controls, rather than to predict FID; Appendix A gives the derivation and its conditions. Crossing the baseline and enlarged target gaps with the two denominators gives Table 1.

At the same state,

$$\ell_D = \ell_A/g, \quad G_D = G_A/g, \quad G_B = G_C/g. \quad (6)$$

Thus D preserves the target-time rule and the relative weighting of different noise levels, while reducing the whole gradient by $1/g$. This is different from choosing a new timestep-weighting function. The native denominator is constructed by subtracting noise times, so a scalar loss wrapper need not reproduce every floating-point operation. Also, loss scaling and learning-rate scaling are distinct: only the former directly changes the gradients entering moment estimation and AMP.

3.2 From one state to an early history

A history applies an operation over many attempts. As states diverge, even the same target-time rule produces different target outputs. We write XY for X during 0–512 king followed by Y during 512–1,024 king. The common A suffix shares the training rule and budget; each branch retains its own parameters, optimizer, scaler, RNG, and data-position state. This design measures whether an early operation leaves an endpoint difference after the rule becomes common. It neither resets the histories to the same model nor directly measures a natural mediator.

The same-state decomposition cannot be summed into an endpoint FID decomposition. Later updates depend on earlier parameter and moment differences, and AMP introduces discrete execution branches. Appendix A.4 formalizes this separation between producing a local difference and propagating an existing state difference. We use the identity to specify experiments and reserve claims about persistent quality for the measured endpoints.

4 A denominator history survives common continuation

4.1 Setting, readout, and cohort relationships

We use unconditional CIFAR-10, an EDM-pretrained DDPM++ network, the stage-zero $q = 256$ mapping, dropout 0.2, and batch size 128 with microbatches of 16 (Karras et al., 2022). Training uses FP16 AMP and RAdam with LR 10^{-4} , betas (0.9, 0.999), $\varepsilon = 10^{-8}$, and zero weight decay; TF32 is disabled. At the 512-king switch, the primary component readout initializes a fresh evaluation EMA, E_{512} , from each branch’s online parameters, then averages its subsequent trajectory under the same rule. This removes the pre-switch EMA

Table 2: Completion and dependence of the principal evidence. Counts refer to training endpoints, not evaluation blocks. “Complete” requires the paths used by that comparison. Quality estimates with missing paths are conditional on endpoint availability.

Study	Planned seeds	Available endpoints	Reuse and interpretation
Joint history	30	AA 26; BA 29; complete 26	Discovery cohort; some controls recur below.
Component history	16	AA 14; BA 15; CA 15; DA 16; complete 14	Reuses AA/BA; adds CA/DA.
Startup pilot	8	All four paths 8; complete 8	Reuses AA/DA; seeds selected using prior response.
Adjacent window	8	AA/early/delayed each 8; complete 8	Adds delayed paths; reuses AA/early on the same eight seeds.
Fresh $q128$	8	All four paths 8; complete 8	Seeds 301–308; four fresh paths, no reused endpoints.
Restore / hold	16	DA 16; DD 16; complete 16	Same D prefixes; DA outcomes reused.
ImageNet boundary	3	IA 3; IB 3; complete 3	Different configuration; 10 checkpoints and 2 NFEs per path.

accumulator as the readout’s initialization requirement, while preserving the source online model and its later influence.

For the component, startup, adjacent-window, and fresh $q128$ studies, three 50k-image evaluation blocks give

$$Y_{s,X} = \frac{1}{3} \sum_{b=0}^2 \log \text{FID}_{s,X,b}, \quad d_s = Y_{s,X} - Y_{s,A}. \quad (7)$$

The independent units are training seeds. We report paired geometric FID ratios $\exp(\bar{d})$ and seed-level t intervals; FID is lower-is-better and KID is auxiliary (Heusel et al., 2017; Bińkowski et al., 2018). The primary endpoint uses E_{512} , FP32 evaluation, and NFE1. The earlier joint-history study retains its own frozen single-block protocol. Table 2 makes reuse and missing outcomes explicit; these studies are not pooled as independent replications.

4.2 The observation and its component test

The joint-history comparison finds an 8.60% geometric FID reduction for BA relative to AA across 26 complete pairs: mean log difference -0.089946 , 95% CI $[-0.129886, -0.050006]$, and 22/26 favorable pairs. Five of 60 planned trajectories have numerical failures. Retrospective source evaluation finds 11 pairs in which B has worse FID at 512 king but better endpoint FID. This motivates component attribution: source quality alone does not determine the subsequent ranking (Appendix C.1).

The component extension adds C/D histories to the existing A/B controls for seeds 50–65. The primary set contains 14 complete four-path seeds, $S_4 = \{50, \dots, 57, 59, \dots, 64\}$. AA is unavailable for 58/65, BA for 58, and CA/58 fails at 971.776 king. No large finite FID is excluded. On the common set, define

$$H_T = Y_{CA} - Y_{AA}, \quad H_W = Y_{DA} - Y_{AA}, \quad I = Y_{BA} - Y_{CA} - Y_{DA} + Y_{AA}. \quad (8)$$

H_T and H_W are simple effects at the baseline level of the other component. We apply Holm correction to these three tests (Holm, 1979); displayed intervals are marginal nominal 95% intervals.

The denominator-only history improves geometric FID by 8.75%, with ratio CI $[0.87334, 0.95349]$ and 13/14 favorable seeds. It can therefore produce a persistent quality benefit while retaining the baseline target-time rule. Target-only history also has a

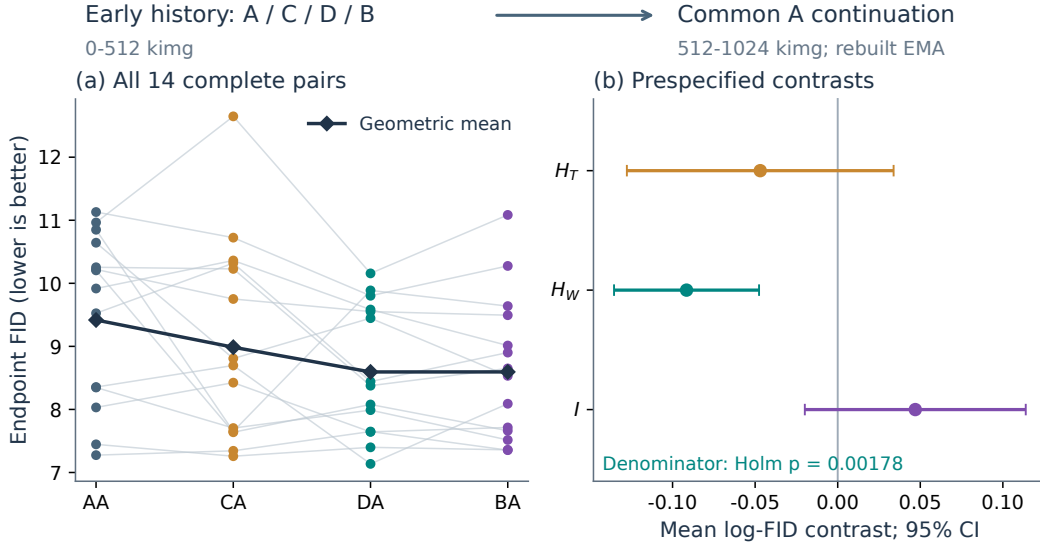


Figure 2: A beneficial history with the baseline target-time rule. Top: each early arm continues under A. Lower left: within-seed geometric FIDs for the 14 complete seeds and their across-seed geometric means. Lower right: paired log-FID contrasts and nominal 95% t intervals. The denominator-only effect H_W passes the three-comparison Holm procedure. The full 16-seed completion counts are in Table 2.

Table 3: Component-history results on the same 14 complete seeds. The upper-panel percentages are descriptive comparisons with AA. Lower-panel tests use seed-level paired log-FID differences.

Path	AA	CA	DA	BA
Geometric FID	9.418	8.987	8.594	8.595
Change vs. AA	—	−4.58%	−8.75%	−8.74%
Contrast	Mean	Seed SD	Nominal 95% CI	Holm p
H_T	−0.04691	0.13987	[−0.12767, 0.03385]	0.30631
H_W	−0.09153	0.07604	[−0.13543, −0.04763]	0.00178
I	0.04697	0.11585	[−0.01991, 0.11386]	0.30631

favorable mean, but its interval is wider. The direct CA–DA interval, [−0.03394, 0.12317] in log-FID, does not establish a ranking between the components. BA and DA have nearly identical means, but their difference interval [−0.03660, 0.03673] does not establish equivalence. Consequently, the factorial comparison identifies an effective operation without allocating a percentage of the original joint benefit to a mediator.

5 Startup update scale and timing

5.1 A known optimizer property specifies the intervention

Why might a uniform loss-scale change survive adaptive normalization? RAdam maintains first and second gradient moments and uses a rectification statistic (Liu et al., 2020):

$$\rho_\infty = \frac{2}{1 - \beta_2} - 1, \quad \rho_n = \rho_\infty - \frac{2n\beta_2^n}{1 - \beta_2^n}. \quad (9)$$

The audited implementation rectifies when $\rho_n > 5$. For $\beta_2 = 0.999$, $\rho_5 \simeq 4.995998$ and $\rho_6 \simeq 5.994164$: its first five successful updates use the unrectified branch. Ignoring weight

Table 4: Completed startup pilot, $n = 8$. M and C are geometric FID changes; S is in log-FID units. Simple-effect p values are Holm adjusted; S uses the primary paired t test. Intervals and tests are nominal after the interim look.

Contrast	Estimate	Nominal 95% CI	p
M_{start}	-12.01%	[-16.77%, -6.97%]	0.001949
C_{start}	+6.04%	[-0.55%, +13.08%]	0.06751
S (log scale)	0.09329	[0.04903, 0.13756]	0.001594

decay and the stabilizing denominator term, its update has the form

$$U_n = \begin{cases} -\eta_n \hat{m}_n, & \rho_n \leq 5, \\ -\eta_n r_n \hat{m}_n / \sqrt{\hat{v}_n}, & \rho_n > 5, \end{cases} \quad (10)$$

where r_n is the rectification factor. If all preceding gradients differ by one fixed positive multiplier s and initial moments scale compatibly, then $m'_n = sm_n$ and $v'_n = s^2 v_n$. The startup update scales by s ; the ideal normalized update cancels it. This conditional scaling property does not imply that diverging networks continue to produce proportional gradients.

We hypothesize that D’s smaller startup steps contribute to its quality advantage. This predicts two endpoint directions: reducing A’s startup LR should improve quality, while compensating D’s startup LR upward should worsen it. The optimizer formula specifies the multipliers and the five-successful-update window; the predicted FID directions additionally require the hypothesis and are tested empirically. Define $A_{\downarrow 5}$ as native A with LR η/g for updates 1–5, and $D_{\uparrow 5}$ as native D with LR $g\eta$ for those updates, followed by A after 512 king. Both restore the original LR from update 6. The interventions do not directly replace gradients, moments, or scaler states; feedback can subsequently change all of them.

Local manipulation check. Two engineering seeds, 50/51, each run 64 attempts. The first common successful update occurs at attempt 9. Native D/A update-norm ratios are 0.909075/0.909062. D compensation gives ratios to A of 0.999983/0.999968; A reduction gives ratios to D of 1.000017/1.000032. Relative vector discrepancies are 0.0195%–0.0235%. This verifies the intended action at the first common state. Later updates include trajectory divergence and possible differences in data position at an equal successful-update index, so the check does not certify five-step trajectory equivalence.

5.2 A completed eight-seed endpoint pilot

The pilot fixes seeds 50–57, reuses their AA/DA controls, and adds two startup trajectories per seed. All 16 new trajectories and 48 new evaluation blocks complete; 48 control blocks are reused. The cohort is responder-enriched, selected using prior response information. Its fixed primary contrast is

$$C_{\text{start}} = Y_{D_{\uparrow 5}} - Y_{DA}, \quad M_{\text{start}} = Y_{A_{\downarrow 5}} - Y_{AA}, \quad (11)$$

$$S = (C_{\text{start}} - M_{\text{start}})/2. \quad (12)$$

The two predictions are $C_{\text{start}} > 0$ and $M_{\text{start}} < 0$. A positive S alone does not establish both. The protocol fixed these contrasts before the endpoint pilot; a six-seed interim view preceded completion, so ordinary intervals and p values remain nominal with respect to repeated inspection. Training, evaluation, and sample size were not adapted to that view.

A startup reduction lowers geometric FID by 12.01%, with all eight seeds improving. D compensation raises it by 6.04%, with six seeds worsening. The latter interval spans a 0.55% improvement to a 13.08% deterioration: it supports the predicted direction but is less precise and does not meet the two-sided Holm threshold. S is positive in all eight seeds; its exact sign-flip sensitivity test gives $p = 0.0078125$. Appendix D reports all seed-level values and the scope of these tests.

In the same cohort, DA improves over AA by 11.28%. The A-reduction/DA ratio is 0.99186 (nominal 95% CI [0.95656, 1.02846]), placing their average qualities close without establishing 3% equivalence. Similar effect sizes do not identify a mediated fraction. The completed

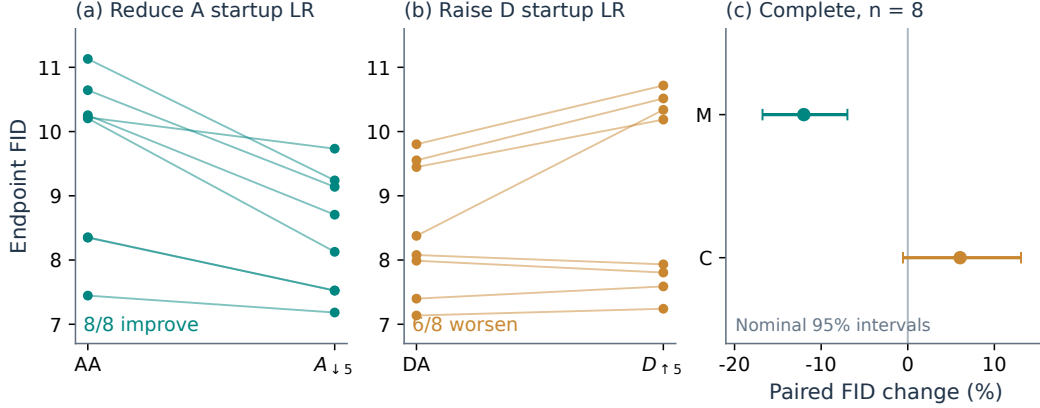


Figure 3: A five-update operation changes final generation quality. The first two panels show all eight paired endpoints: A startup reduction improves 8/8 seeds, while D compensation worsens 6/8. The third panel shows the two simple effects with nominal 95% paired-t intervals. The cohort was selected using prior response information and reuses controls. Intervals are nominal after an earlier six-seed inspection.

experiment shows that a short update operation has a lasting endpoint consequence; its reverse arm gives additional directional support. Together with the component history, it motivates startup scale as a candidate pathway for the original effect. Proving that this pathway carries the full benefit would require further interventions that distinguish it from later parameter, moment, and numerical feedback.

5.3 Moving the same reduction to the next five updates

The startup result motivates a direct timing question: does lowering LR for five updates have a comparable effect immediately after the unrectified branch? We add $A_{\downarrow 6:10}$, native A with LR η/g on successful updates 6–10, restoring the base LR from update 11. The early arm $A_{\downarrow 5}$ retains its updates 1–5 reduction. Each delayed path starts from its seed’s original transfer initialization, runs for the same 1,024-king budget, and initializes its own E_{512} once at 512 king. No optimizer restart occurs.

All eight new trajectories and 24 new evaluation blocks complete. We exactly reuse the 48 AA/early blocks, giving three complete paths for each of seeds 50–57. The extension uses the same responder-enriched cohort and was motivated by the completed startup pilot. Its new quality results were unblinded after the full group completed; this does not make it a fresh independent confirmation. The frozen primary comparison and the secondary baseline comparison are

$$T_s = Y_{s,A_{\downarrow 6:10}} - Y_{s,A_{\downarrow 5}}, \quad L_s = Y_{s,A_{\downarrow 6:10}} - Y_{s,AA}, \quad T_s = L_s - M_{\text{start},s}. \quad (13)$$

The identity makes the reuse explicit: the previously measured M_{start} is context, not an additional replication.

The direct comparison gives $\bar{T} = 0.106596$, nominal 95% log interval $[0.042848, 0.170344]$, and two-sided paired $p = 0.005504$. Delayed/early geometric FID is 1.11248, or 11.25% higher (95% interval $[4.38\%, 18.57\%]$), with all eight seed differences positive. Exhaustive sign-flip sensitivity gives $p = 0.0078125$. Delayed/AA has a smaller estimated improvement of 2.11%, with interval $[-4.26\%, +0.10\%]$ and $p = 0.05817$. This does not establish that delaying has no effect. Evidence for a window difference comes from T , not from comparing the significance labels of two baseline tests. Appendix D.1 reports every seed and all preset comparisons.

What the timing contrast resolves. The two placements are not interchangeable in this cohort. This strengthens the account from “a short LR reduction helps” to a measured

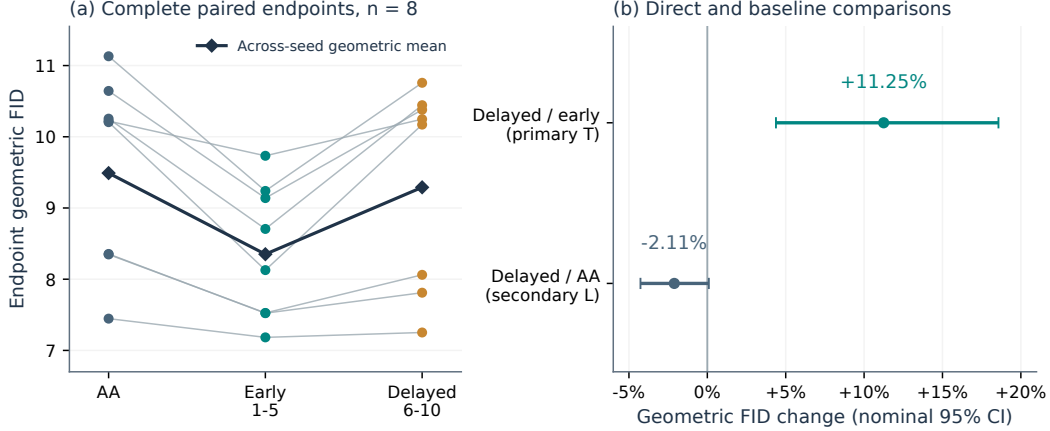


Figure 4: The same five-update LR reduction has different endpoint effects in adjacent windows. Left: all eight paired geometric FIDs under AA, early reduction (successful updates 1–5), and delayed reduction (6–10). AA and early outcomes are reused. Right: the direct delayed/early comparison T and secondary delayed/AA comparison L , exponentiated to FID percentage changes with nominal 95% paired-t intervals. All eight seeds favor early over delayed. Both interventions divide LR by 1.1 for five successful updates; their realized update exposure differs (Figure 7).

dependence on where that operation is applied. It does not isolate time from the rest of the update process. Across all 16 early/delayed trajectories, the sum of actual update norms in updates 1–5 is 22.22–41.82 times its value in updates 6–10. These sums describe realized movement, not the counterfactual perturbation introduced by the LR change. Equal update count and LR multiplier therefore do not equalize actual movement or intervention dose. The branch transition, moment history, numerical execution, and their later feedback remain possible contributors. A unique RAdam mediator, a dose-matched critical-period claim, and the fraction of the natural denominator-history benefit explained by startup scale remain unestablished.

5.4 A fresh q128 cohort limits transfer of the startup response

The new cohort fixes seeds 301–308 without selecting them for a previous quality response. Each seed has four fresh trajectories: AA, DA, $A_{\downarrow 5}$, and $D_{\uparrow 5}$. Native $q = 128$ is retained before and after 512 king; target and global scales remain one. DA and $D_{\uparrow 5}$ use denominator scale 1.1 only in the prefix, then return to native A. Each path retains its own training state and initializes its own E_{512} at the boundary. All 32 trajectories and 96 evaluation blocks complete, yielding eight complete training-seed units.

The prespecified primary contrast remains $S = (C - M)/2$, with $M = Y_{A_{\downarrow 5}} - Y_{AA}$ and $C = Y_{D_{\uparrow 5}} - Y_{DA}$; $H = Y_{DA} - Y_{AA}$ is supportive. The primary estimate is $\hat{S} = 0.004235$, nominal 95% paired-t CI $[-0.034159, 0.042629]$, $p = 0.801741$ (Table 5). Four seeds have positive S and four negative. The geometric FID changes corresponding to M , C , and H are +0.47%, +1.32%, and −0.89%, respectively. Both M/C Holm-adjusted p values are 1.0. The cohort does not reproduce the expected A-side startup benefit or provide stable support for the denominator-history advantage. These intervals do not establish a zero effect or equivalence.

A bounded audit of existing records confirms the operation: 763 continuous attempts cover the first 16 successful updates of all 32 trajectories. The observed LR windows, restoration, scaler events, and successful-call clock checks match the plan. Relative to each seed’s own AA path, $A_{\downarrow 5}$ reduces the sum of the first five update norms in all eight seeds (ratios 0.768–0.957). Relative to DA, $D_{\uparrow 5}$ increases it in all eight (1.005–1.269). Recorded net displacement is smaller in all eight A-reduction pairs and larger in seven of eight D-compensation

Table 5: Fresh q128 results, $n = 8$ training seeds. All estimates and intervals are in log-FID units. S is primary, M/C are the two Holm-adjusted secondary tests, and H is supportive. Intervals are nominal 95% paired-t intervals, not simultaneous intervals. $\exp(S)$ is a composite ratio, not a model’s percentage FID improvement.

Contrast	Mean	Nominal 95% CI	Raw p	Holm p
S	+0.004235	$[-0.034159, +0.042629]$	0.801741	—
M	+0.004645	$[-0.030614, +0.039903]$	0.764503	1.000000
C	+0.013114	$[-0.037362, +0.063590]$	0.558415	1.000000
H	-0.008912	$[-0.054840, +0.037016]$	0.660248	—

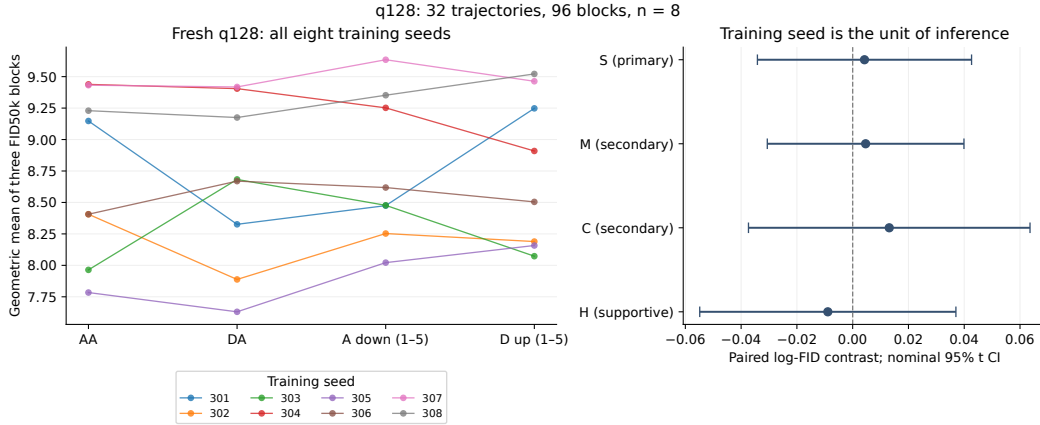


Figure 5: The fresh q128 cohort does not reproduce a stable endpoint benefit. Left: all four endpoints for every prespecified seed; each point is the geometric mean of three FID50k blocks. Right: the prespecified seed-level contrasts and nominal 95% paired-t intervals. The blocks are repeated evaluations of a trained model, not independent training replicates or FID150k.

pairs. These are comparisons of evolving trajectories, not same-state counterfactual perturbation doses. The adjacent-window readings and complete per-seed values are retained in Appendix G.

The local response therefore occurred without stable evidence of the predicted endpoint improvement. The existing material does not explain that discrepancy. Comparing these results with the responder-enriched $q256$ studies changes both q and cohort, so it does not identify a causal effect of q or an explained condition under which the mechanism succeeds or fails.

6 State interventions, scope, and remaining questions

Restart and swap answer different questions. M1 crosses A/B history with optimizer retention or restart. Keep-A/B complete 14/16 and 15/16 paths; restart-A/B complete 8/16 and 15/16. The restart-history comparison has seven complete pairs and remains uncertain. A local decomposition on four retrospectively selected seeds shows that clearing moments at a fixed clock can enlarge the next update norm by 12.7–14.7 times. Restart thus changes the update process, beyond deleting a putative historical carrier. Six-seed optimizer swaps and a 512-versus-768-king restart comparison do not identify a unique optimizer-source effect or a memory-decay law (Appendix E). These observations motivate direct update interventions while leaving optimizer mediation unresolved.

A persistent effect does not imply a benefit from withdrawal. From the same D prefix, DA/DD has geometric FID ratio 1.00889 across all 16 pairs (95% CI $[0.97817, 1.04056]$).

Neither the difference test ($p = 0.551$) nor the prespecified 3% equivalence test ($p = 0.0869$) passes. The data therefore do not establish that restoring A at 512 king improves over maintaining D.

Cross-configuration evidence bounds the claim. ImageNet- 64×64 provides three IA/IB pairs, ten checkpoints, and two NFEs. Through 6,400 king, all 30 paired FID readouts favor baseline IA over the enlarged-gap IB; over the complete trajectory, 50/60 do so, with heterogeneous late deterioration retained. This setting uses $c = 0.06$ and gap-independent snrpk weighting, together with different data, architecture, and other settings. It is a boundary on an unconditional gap-improvement claim, not a matched factorial test of the CIFAR mechanism. Appendix F shows all trajectories, along with unresolved or protocol-limited earlier $q128$ and other history studies.

What the remaining tests can establish. The completed window comparison establishes a placement-dependent endpoint response within the existing $q256$ cohort, with unequal realized movement. The fresh $q128$ cohort now shows that correct execution and the expected direction of local movement are insufficient to secure a stable endpoint benefit in another setting and cohort. The data do not explain which aspect of this change matters. A causal role for spacing, a dose-matched timing effect, and a natural mediation fraction remain unresolved. No further training, sample generation, or optimizer/precision search was performed for this closeout.

7 Related work and interpretation

Consistency training and pairing. CMs and improved consistency training establish the consistency objective, single-sided targets, and robust-loss design (Song et al., 2023; Song & Dhariwal, 2024). ECT develops diffusion pretraining followed by tuning and explicitly examines mapping and weighting choices (Geng et al., 2025). TCM changes the noise-time range assigned to consistency training (Lee et al., 2024); ADCM selects discretization through a local/global consistency trade-off and adds adaptive loss components (Bai et al., 2025). Our experiments ask which separately executed component of a gap change can create a benefit that persists under a shared later rule. They do not establish that these newer methods inherit the same effect.

Relative weighting, global scale, and stabilization. Min-SNR- γ treats diffusion timesteps as tasks and balances their optimization conflicts with SNR-dependent weights (Hang et al., 2023). Such changes alter relative emphasis across noise levels. D instead multiplies the complete same-state gradient by $1/g$ under the studied objective. Its effect raises a question about stateful execution that relative reweighting alone does not answer. The sCM/TrigFlow framework improves stability using parameterization, tangent normalization, adaptive weighting, and tangent warmup (Lu & Song, 2025). In particular, its warmup demonstrates that startup stabilization is already part of CM methodology. Our experiments isolate a denominator operation, specify a five-update test from the audited RAdam branch, directly compare early and delayed placement, and test transfer in a fresh cohort. Their contribution is this chain of measured contrasts, rather than the general idea of warmup.

Mechanistic studies of CM learning. Stable Consistency Tuning (SCT) relates consistency learning to temporal-difference learning and reduces variance using a score identity (Wang et al., 2024). Align Your Tangent (AYT) analyzes oscillatory output tangents and changes the feature-space distance to improve their alignment (Kim et al., 2025). These works investigate training behavior and use their analyses to guide modifications. Our local identity separates target-induced direction changes from a uniform scale change; our endpoint experiments then examine history and startup effects. Output tangents along a noise trajectory, parameter-space gradients, and optimizer parameter steps are distinct objects and should not be conflated.

Adaptive optimizers and early training. Adam’s ideal scale cancellation and RAdam’s startup rectification explain why loss scale need not translate uniformly into update scale (Kingma & Ba, 2015; Liu et al., 2020). Finite stabilizers, nonproportional moment histories, and numerical execution qualify that cancellation; Appendix A states the assumptions explicitly. Slingshot documents late-training dynamics in adaptive optimizers (Thilak et al., 2022), while critical-learning-period studies show that early perturbations can have persistent consequences (Achille et al., 2019). These are relevant precedents for studying training paths, rather than evidence that our setting exhibits the same mechanisms. We have not identified a Slingshot process or a uniquely critical learning period.

Evidence strength and reproducibility. Cohort reuse preserves useful within-seed comparisons but prevents treating every study as an independent replication. Differential failures restrict complete-case quality inference; transparent accounting does not remove that selection. The responder-enriched startup cohort and its interim inspection further limit population claims. The adjacent-window extension adds new delayed trajectories but shares that cohort and its AA/early endpoints; full-group unblinding of the extension does not remove the earlier selection. The accompanying artifact provides the measurements, source hashes, and code needed to reconstruct figures and summary statistics. It does not include all training data, model checkpoints, or private historical execution receipts; reproducing the full training campaign additionally requires those assets. Appendix D.3 distinguishes verified public result bindings from remaining execution-archive gaps. The new q128 cohort narrows the transfer claim: the earlier operation-level effects remain measured results in their own cohorts, while a broadly transferable mechanism remains unestablished.

8 Conclusion

Separating target construction from loss scale shows that an early denominator-only operation can improve consistency-tuning quality after common continuation in the studied $q256$ setting. Startup and adjacent-window interventions connect this observation to a known optimizer response and reveal placement-dependent endpoint effects in a responder-enriched cohort. The fresh $q128$ cohort does not reproduce a stable startup or denominator-history benefit, despite verified execution and observable local update changes. The resulting contribution is an experimentally delimited account of loss-scale history, not a general mechanism inferred from a local identity. Existing records leave the difference between local response and endpoint benefit unexplained.

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A Mathematical details and interpretation limits

A.1 Consistency paths and the training coupling

For the variance-exploding parameterization $x_t = x_0 + t\epsilon$, the probability-flow ODE can be written as

$$\frac{dx_t}{dt} = -t\nabla_x \log p_t(x_t). \quad (14)$$

An ideal consistency function is constant along each ODE trajectory and satisfies the corresponding data-end boundary condition (Song et al., 2021; 2023). Along a smooth trajectory,

$$\frac{d}{dt}f(x_t, t) = \partial_t f(x_t, t) + D_x f(x_t, t) \frac{dx_t}{dt} = 0. \quad (15)$$

In the shared-noise training construction, the samplewise tangent is instead ϵ . Its conditional mean satisfies

$$\mathbb{E}[\epsilon \mid x_t] = -t\nabla_x \log p_t(x_t), \quad (16)$$

using the conditional score identity for Gaussian perturbations. Equality after conditional expectation does not make each shared-noise line an exact ODE path. Our experiments change the training coupling and its weight; they do not intervene on a sampling solver’s discretization.

A.2 Proof of the finite-gap decomposition

For fixed student input and random realization, differentiation of Equation 3 through the student alone gives

$$\nabla_{\theta}^{\text{sg}} \ell(r, \delta) = \frac{1}{\delta} J_t^{\top} \frac{e_r}{\|e_r\|_2}. \quad (17)$$

Substituting $s = \delta_1/\delta_g$ and adding and subtracting the baseline residual direction proves Lemma 1. The associated forward-value identity is

$$\ell(r_g, \delta_g) = s\ell(r_1, \delta_1) + \frac{\|e_{r_g}\|_2 - \|e_{r_1}\|_2}{\delta_g}. \quad (18)$$

The field identity compares the same student Jacobian. It requires finite nonzero residuals; the Euclidean norm at zero requires a specified subgradient convention and is outside the stated lemma. The $c > 0$ pseudo-Huber case uses a different residual-direction factor.

For the four arms, $G_D = sG_A$ and $G_B = sG_C$ under the stated conditions, giving

$$G_B - G_C - G_D + G_A = (s - 1)(G_C - G_A). \quad (19)$$

This is a same-state factorial identity. The endpoint interaction I is evaluated after independently evolving trajectories and need not satisfy an analogous gradient-based formula. In particular, its sign cannot be inferred from the factor $s - 1$ alone.

A.3 Why stop-gradient changes an asymptotic argument

Introduce a two-parameter residual

$$e_{\delta}(\theta_s, \theta_t) = f_{\theta_s}(x_t, t) - f_{\theta_t}(x_{t-\delta}, t - \delta). \quad (20)$$

The forward value is evaluated on the diagonal (θ, θ) , but the training derivative acts only on θ_s . Suppose

$$e_{\delta}(\theta, \theta) = \delta^{\nu} v(\theta) + O(\delta^{\nu+1}), \quad w(\delta) \sim w_0 \delta^{-\alpha}. \quad (21)$$

Here $\delta \downarrow 0$, $\nu > 0$, $w_0 > 0$, and $v(\theta) \neq 0$. Let the outer loss φ be locally homogeneous of degree p_{hom} , with the differentiability and nondegeneracy conditions needed for its leading derivative. If the student Jacobian converges and its projection onto $\nabla \varphi(v)$ is nonzero, the leading single-sided field scales as

$$\nabla_{\theta_s} [w(\delta) \varphi(e_{\delta})]_{\theta_s = \theta_t = \theta} \sim \delta^{\nu(p_{\text{hom}} - 1) - \alpha} w_0 J_0^{\top} \nabla \varphi(v). \quad (22)$$

This follows from $\nabla\varphi(\delta^\nu v) = \delta^{\nu(p_{\text{hom}}-1)}\nabla\varphi(v)$ and the nonvanishing Jacobian limit. Differentiating the diagonal expansion as a two-sided forward function would instead give exponent $\nu p_{\text{hom}} - \alpha$, provided its remainder is uniformly differentiable in the parameters. That extra assumption does not follow from a pointwise residual expansion.

For the norm-over-gap objective, $p_{\text{hom}} = 1$ and $\alpha = 1$, so Equation 22 gives exponent -1 under the stated nondegeneracy assumptions. This is a conditional local statement, not a universal divergence theorem for trained networks. A scalar example makes the derivative distinction explicit: for $f_\theta(s) = \theta s$, $t = 1$, $r = 1 - \delta$, the diagonal forward loss is $|\theta|$, with ordinary derivative $\text{sign}(\theta)$; its single-sided derivative is $\text{sign}(\theta)/\delta$ for $\theta \neq 0$.

A.4 State differences and a conditional propagation bound

Let Z_k collect the full resumable state: parameters, optimizer and evaluation-EMA states, scalar state, random generators, and data position. Write Φ_X for one native attempt under operation X . On compatible state coordinates, adding and subtracting $\Phi_A(Z_k^B)$ gives

$$Z_{k+1}^B - Z_{k+1}^A = \underbrace{\Phi_A(Z_k^B) - \Phi_A(Z_k^A)}_{\text{propagation of different states}} + \underbrace{\Phi_B(Z_k^B) - \Phi_A(Z_k^B)}_{\text{same-state operation difference}}. \quad (23)$$

The first term includes feedback through subsequent targets and parameter updates. This bookkeeping identity separates two questions; it does not identify a carrier or explain the sign of a quality change. Equation 23 remains an algebraic identity without a smoothness assumption. A useful bound requires more. Restrict attention to compatible continuous state coordinates, a fixed external random input sequence, and a region in which the reference transition is L_k -Lipschitz. Let b_k denote the same-state operation difference. Then

$$\|\Delta Z_{k+1}\| \leq L_k \|\Delta Z_k\| + \|b_k\|, \quad (24)$$

and iteration yields

$$\|\Delta Z_K\| \leq \left(\prod_{j=0}^{K-1} L_j \right) \|\Delta Z_0\| + \sum_{i=0}^{K-1} \left(\prod_{j=i+1}^{K-1} L_j \right) \|b_i\|. \quad (25)$$

Empty products are one. This inequality illustrates how local differences can be propagated or contracted. It neither determines the realized signs of state differences nor the direction of FID. Random-generator state and discrete AMP branches cannot silently be treated as globally differentiable coordinates. We do not estimate a validated sequence of L_k or claim a global contraction property.

A.5 RAdam scaling and its execution limits

With G_n the gradient supplied on a successful update,

$$m_n = \beta_1 m_{n-1} + (1 - \beta_1) G_n, \quad v_n = \beta_2 v_{n-1} + (1 - \beta_2) G_n^{\odot 2}. \quad (26)$$

If $G'_n = s G_n$ for every previous successful update, $m'_0 = s m_0$, $v'_0 = s^2 v_0$, and clocks coincide, induction gives $m'_n = s m_n$ and $v'_n = s^2 v_n$. Bias corrections preserve these relations. Unrectified updates then scale by s . In the ideal normalized branch with a zero stabilizing term, on coordinates with positive second moment, $s \hat{m}_n / \sqrt{s^2 \hat{v}_n} = \hat{m}_n / \sqrt{\hat{v}_n}$ for $s > 0$.

The actual optimizer uses $\varepsilon = 10^{-8}$ and finite arithmetic, and scaled startup updates change the later network state. The required proportional-gradient history therefore need not continue. AMP can also change which attempts produce updates. These qualifications are the reason for both a local manipulation check and a full endpoint test. Compensating the LR does not overwrite moments with those from the comparator trajectory.

Why a finite stabilizer matters. For the schematic Adam normalization $U = -\eta \hat{m} / (\sqrt{\hat{v}} + \varepsilon)$ and a positive scale s , proportional moments give

$$U' = -\eta \frac{s \hat{m}}{s \sqrt{\hat{v}} + \varepsilon} = -\eta \frac{\hat{m}}{\sqrt{\hat{v}} + \varepsilon/s}. \quad (27)$$

Thus a fixed nonzero stabilizer breaks exact cancellation; its magnitude relative to the second moment determines the discrepancy. The precise placement of the stabilizer and bias corrections is implementation dependent. This illustration is not a measurement that ε drives the observed FID effect. Moreover, cancellation requires proportional histories, not merely two current gradients related by a scalar. Once startup steps separate the networks, this requirement can fail even in exact arithmetic. Our present intervention therefore targets a concrete update operation while leaving the relative contributions of moment transients, finite stabilizers, and AMP unresolved.

A.6 Spacing parameters and floating-point equivalence

In the audited stage-zero mapping, $n(t) = 1 + 8/(1 + \exp(t))$ and the unclipped baseline relative gap is $n(t)/q$. For $t > 0$, $1 < n(t) < 5$, and at $q = 256$ the relative gap lies between approximately 0.39% and 1.95%. Multiplying the gap by g changes the real-arithmetic ratio to $gn(t)/q$. Thus changing (q, g) while fixing g/q does not create a new independent pair geometry in this stage. The general code uses a stage exponent; the same statement must be re-evaluated if that stage changes.

Recomputing a time and subtracting it from t can round differently from multiplying an already computed gap. The included analysis treats real identities as definitions of intervention meaning, while preserving the native execution path in training. Local numerical agreement is reported only at the tested states and precisions, rather than promoted to whole-trajectory equivalence.

B Protocol, readouts, and data provenance

Table 6: Main component-history and startup configuration.

Item	Setting
Data / transfer	CIFAR-10 32×32 ; native EDM-pretrained unconditional transfer
Network	DDPM++; ECT preconditioning; dropout 0.2
Mapping / loss	Sigmoid; $q = 256$, $k = 8$, $b = 1$, $c = 0$; stage remains zero
Batching	Global batch 128; microbatch 16; one GPU per trajectory
Optimizer	RAdam; LR 10^{-4} ; betas (0.9, 0.999); eps 10^{-8} ; weight decay 0
Precision	Native FP16 + AMP training; TF32 disabled; FP32 quality evaluation
Augmentation	No augmentation; no x-flip
Noise sampling	Lognormal parameters mean -1.1 , standard deviation 2.0
Budget / boundary	1,024 king; switch at 512 king; attempted images count toward budget
Evaluation EMA	Beta 0.9993; E_{512} initialized once from online weights at 512 king
Endpoint blocks	NFE1; sample seeds 0–49,999, 50,000–99,999, and 100,000–149,999

All continuation experiments preserve resumable training state except for the operation explicitly under test. Normal AMP skips advance the attempt and data counters but not RAdam’s successful-update clock; no extra examples are supplied to equalize successful-update counts. Nonfinite scientific failures are recorded under the original stopping rules. The component study retains native denominator construction rather than substituting a scalar loss wrapper.

Scope of pairing checks. The component extension verifies initial states and available prefix telemetry, including batch, labels where applicable, t , and base r , against the original controls. The available record does not provide per-attempt hashes for every historical epsilon and dropout realization. Short resume checks support the implementation paths they cover; they do not establish a full 8,000-attempt bitwise replay of every old and new trajectory. Controls were reused across execution periods, which is a limitation in addition to cohort reuse.

Evaluation EMA. If E_{512} starts from θ_{512} and updates once per attempt with coefficient β , then after N suffix attempts

$$E_N = \beta^N \theta_{512} + (1 - \beta) \sum_{j=1}^N \beta^{N-j} \theta_j. \quad (28)$$

For $\beta = 0.9993$ and $N = 4000$, the direct initialization coefficient is about 0.061. Rebuilding the EMA does not erase the source online weights or the influence they exert through later trajectories. The teacher used in Equation 3 is not this evaluation EMA.

Artifact provenance. The accompanying source manifest maps each plotted CSV and archived statistic to a repository commit, original path, and SHA-256 hash. Main snapshots include the joint-history study (890a85a8), source backfill (200739fa), component extension (90a0cd3f), restore/hold study (092ab9ab), longitudinal study (d7084c7f), completed startup pilot (e429ab29), and adjacent-window extension (3d1330f5). The startup figures now use full-precision block FIDs and the final eight-seed table; the earlier screenshot transcription is archived separately and excluded from current calculations. The artifact contains measurements and plotting code, not model checkpoints or generated samples.

C Completion accounting and seed-level component results

The component cohort plans all four paths for seeds 50–65. AA completes for 14 seeds, BA for 15, CA for 15, and DA for all 16. Seed 58 lacks AA, BA, and CA; seed 65 lacks AA. All other planned paths complete. The 14-seed complete set excludes neither a large finite FID nor a seed on the basis of its contrast. CA/58 fails near the endpoint, after 7,576 successful updates and 16 skips, at 971.776 king. Failure does not justify substituting a different-precision rescue into the primary analysis.

Table 7: All complete component seeds. Cells are geometric means of the three per-seed FIDs; H_W is computed on the original log values.

Seed	AA	CA	DA	BA	H_W
50	11.130	10.724	9.803	10.276	-0.12704
51	8.355	8.697	7.137	8.090	-0.15759
52	8.348	7.705	7.989	7.516	-0.04395
53	10.205	7.638	8.078	7.662	-0.23373
54	7.446	7.258	7.399	7.359	-0.00638
55	10.220	9.751	9.552	9.494	-0.06761
56	10.644	8.808	9.447	8.533	-0.11925
57	10.254	10.228	8.376	8.650	-0.20227
59	7.275	7.344	7.649	7.354	+0.05013
60	9.525	10.320	8.442	8.902	-0.12066
61	8.032	8.425	7.644	7.714	-0.04952
62	9.918	10.362	9.584	9.013	-0.03420
63	10.965	12.647	10.157	11.085	-0.07647
64	10.849	7.652	9.887	9.639	-0.09287

The 30-seed history study, M1 keep controls, the component extension, restore/hold study, startup pilot, and adjacent-window extension overlap in seeds and/or checkpoints. Repeated readouts and additional interventions do not create additional independent realizations of the original A/B history. Completion rates are reported alongside conditional quality effects; an unconditional effect over failed and successful runs would require an explicitly defined outcome for failure or a different estimand.

C.1 Joint-history observation and retrospective source evaluation

The source-to-future comparison uses 29 valid source pairs at 512 king, 26 of which also have both endpoints. Of 13 pairs with worse B source FID in the common set, 11 have better

Table 8: Availability for all 16 planned component seeds. Yes means a recorded usable endpoint; dashes mark missing or failed outcomes. The primary quality contrast uses only the common four-path set, without imputing an FID for failure.

Seed	AA	BA	CA	DA	In primary set
50	Yes	Yes	Yes	Yes	Yes
51	Yes	Yes	Yes	Yes	Yes
52	Yes	Yes	Yes	Yes	Yes
53	Yes	Yes	Yes	Yes	Yes
54	Yes	Yes	Yes	Yes	Yes
55	Yes	Yes	Yes	Yes	Yes
56	Yes	Yes	Yes	Yes	Yes
57	Yes	Yes	Yes	Yes	Yes
58	—	—	—	Yes	—
59	Yes	Yes	Yes	Yes	Yes
60	Yes	Yes	Yes	Yes	Yes
61	Yes	Yes	Yes	Yes	Yes
62	Yes	Yes	Yes	Yes	Yes
63	Yes	Yes	Yes	Yes	Yes
64	Yes	Yes	Yes	Yes	Yes
65	—	Yes	Yes	Yes	—

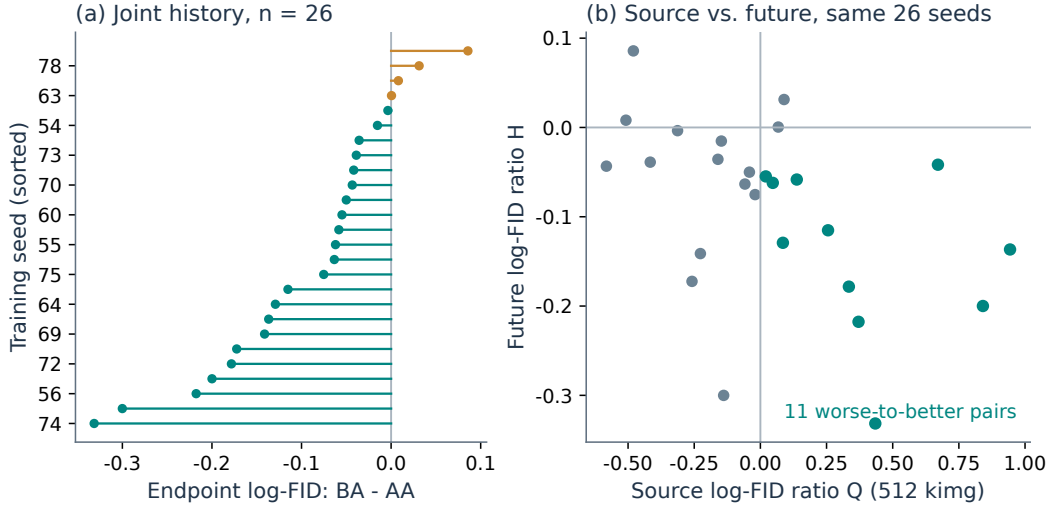


Figure 6: Earlier quality does not fully determine later relative quality. Left: all 26 complete AA/BA endpoint log-FID contrasts. Right: retrospective source contrast $Q = \log(\text{FID}_{B,512} / \text{FID}_{A,512})$ versus the endpoint contrast H . Eleven common pairs have $Q > 0$ and $H < 0$. Both panels use the same endpoint cohort.

B-history endpoint FID. Figure 6 shows the paired observations; they reuse the joint-history endpoint cohort and do not constitute an independent replication.

D Statistical definitions and startup observations

For complete seed-level differences d_s , our displayed nominal 95% t interval is

$$\bar{d} \pm t_{n-1,0.975} s_d / \sqrt{n}. \quad (29)$$

Exponentiating gives an interval for the geometric FID ratio. This is not the same as a ratio of arithmetic FID means or an FID computed from pooling 150k images. The component

family’s three tests use Holm correction at 0.05; these adjusted decisions do not turn the displayed marginal intervals into simultaneous intervals.

To assess a multiplicative 3% equivalence margin, the log bounds are $[-\log(1.03), \log(1.03)]$. At level 0.05, the two one-sided tests require the corresponding 90% interval to lie strictly within those bounds (Schuirmann, 1987). A nonsignificant difference with a wide interval is not equivalence. For DA–DD, both the difference test and the equivalence decision remain unresolved at their stated thresholds.

Table 9: All eight startup seeds from the final full-precision results. Cells show geometric means over three evaluation blocks, rounded only for display. All statistical calculations use the unrounded block values.

Seed	AA	$A_{\downarrow 5}$	DA	$D_{\uparrow 5}$
50	11.130	9.239	9.803	10.717
51	8.355	7.523	7.137	7.242
52	8.348	7.526	7.989	7.805
53	10.205	8.128	8.078	7.933
54	7.446	7.183	7.399	7.589
55	10.220	9.733	9.552	10.515
56	10.644	9.139	9.447	10.186
57	10.254	8.706	8.376	10.338

Recomputation from the 96 block FIDs (48 new and 48 reused controls) gives $\bar{M} = -0.127901$, $\bar{C} = 0.058688$, and $\bar{S} = 0.093295$. The corresponding nominal 95% log-FID intervals are $[-0.183576, -0.072227]$, $[-0.005530, 0.122907]$, and $[0.049032, 0.137558]$. Exact paired sign-flip sensitivity p values for M, C, S are 0.0078125, 0.0703125, and 0.0078125. These enumerate all 2^8 sign patterns and rely on the relevant sign-exchangeability assumptions; they do not remove selection or interim-inspection effects. None of the protocol’s three contrasts passes the prespecified equivalence test.

The final matrix contains all 16 planned new trajectories, 48 successful new evaluations, and eight finite complete seed pairs. Block values reconcile with the published new evaluation records and old-control bindings; new checkpoint identifiers match the final training outcomes. This is a numerical reconciliation of the public records, not a rerun of training or an independent rehash of all private historical model and feature archives.

The cohort was selected using prior response information, and a six-seed interim view preceded completion of the fixed eight-seed plan. Training, evaluation, and sample size were not adapted to that view. Holm adjustment covers only the two simple effects, and ordinary intervals and p values remain nominal with respect to repeated looks. Completing seeds 56–57 completes this exploratory pilot; it does not create fresh unselected replication. Directional support for D compensation should therefore be reported with its interval and selection scope, rather than reduced to either proof or absence of an effect.

D.1 Adjacent-window protocol and complete results

The delayed-window extension uses the same $q = 256$, $g = 1.1$, $c = 0$, native-A loss and mapping, optimizer, precision, and evaluator as the startup pilot. Its only prescribed update modification divides LR by 1.1 on successful updates 6–10. The controller advances only after an actual optimizer step; AMP skips do not consume a successful-update position. After update 10 the controller is inactive, with the native LR restored. Every delayed trajectory begins at transfer initialization, completes 8,000 attempted iterations (1,024 king), and rebuilds its own evaluation EMA exactly once at attempt 4,000. This is an independent training path from the matched seed initialization, not a branch resumed after observing its early quality.

The plan fixes eight seeds, 50–57, and uses the complete three-path set. Eight new delayed trajectories and all 24 new FID50k/KID evaluation blocks complete. Exactly 48 AA/early control blocks are reused; the combined 72 blocks represent eight training-seed comparisons,

not 72 independent observations or FID150k. The analysis was unblinded after the complete new group was available. The design is an outcome-informed extension of the responder-enriched pilot. The original pilot’s six-seed interim inspection and the reuse of its endpoints are disclosed; no new unselected cohort is created by adding the delayed arm.

Table 10: All eight adjacent-window seeds. Endpoint cells are geometric FID over three blocks; T_s is delayed minus early in log-FID units. AA and early cells are exact reuse of the previous pilot, rounded here only for display.

Seed	AA	Early	$A_{\downarrow 5}$	Delayed	$A_{\downarrow 6:10}$	T_s
50	11.130		9.239		10.758	0.152169
51	8.355		7.523		7.811	0.037533
52	8.348		7.526		8.061	0.068699
53	10.205		8.128		10.171	0.224232
54	7.446		7.183		7.252	0.009488
55	10.220		9.733		10.247	0.051445
56	10.644		9.139		10.379	0.127228
57	10.254		8.706		10.443	0.181973

Table 11: All preset window comparisons, $n = 8$. Differences and intervals are on the log-FID scale. Tests are nominal two-sided paired t tests. T is the primary direct comparison, L the secondary baseline contrast, and M is a previously measured contextual effect. Its raw p differs from the pilot’s two-simple-effect Holm-adjusted value.

Contrast	Mean	Nominal 95% CI	Raw p
T : delayed – early	0.106596	[0.042848, 0.170344]	0.005504
L : delayed – AA	−0.021305	[−0.043578, 0.000967]	0.058167
M : early – AA (reused)	−0.127901	[−0.183576, −0.072227]	0.000974

Exponentiating $\bar{T}$ gives delayed/early ratio 1.112485 with 95% interval [1.043779, 1.185713]. The delayed/AA ratio is 0.978920, interval [0.957358, 1.000968]; six delayed seeds improve over AA and two worsen. The reused early/AA ratio is 0.879940, interval [0.832289, 0.930320]. The difference in window effects is tested directly by $T = L - M$; different significance labels for L and M would not establish this difference. The raw tests shown here are not a correction for all outcome-informed questions in the research program.

For T , enumerating all 256 sign patterns gives a two-sided sign-flip sensitivity $p = 0.0078125$. This calculation assumes symmetry of the paired differences under the null; it is not a randomization test arising from random arm assignment. An auxiliary TOST at $\pm \log(1.03)$ gives $p = 0.987789$, failing the equivalence criterion. The difference evidence and the failed equivalence check should be reported with their distinct purposes.

D.2 Observed update exposure and its limits

The archive binds the first 16 successful updates of all 16 early/delayed paths to immutable attempt-level prefixes. The 406 retained attempt rows contain 256 successful updates and 150 skipped attempts. Every path applies its reduction on exactly five successful updates at the assigned positions. Figure 7 shows the successful-update LR and norm traces, the rectification statistic, and the scaler/skip records. A successful-update plot alone would hide the attempts on which AMP prevented a parameter update.

For a window W , let $E_W = \sum_{n \in W} \|\theta_n - \theta_{n-1}\|_2$ be the sum of actual update norms on one trajectory. Table 12 reports these sums. Across the 16 paths, $E_{1:5}/E_{6:10}$ ranges from 22.22 to 41.82. These are descriptive realized movements; they do not measure how far an intervened path moves relative to a counterfactual path without the LR reduction. A shared LR factor therefore cannot be read as equal perturbation dose across the windows.

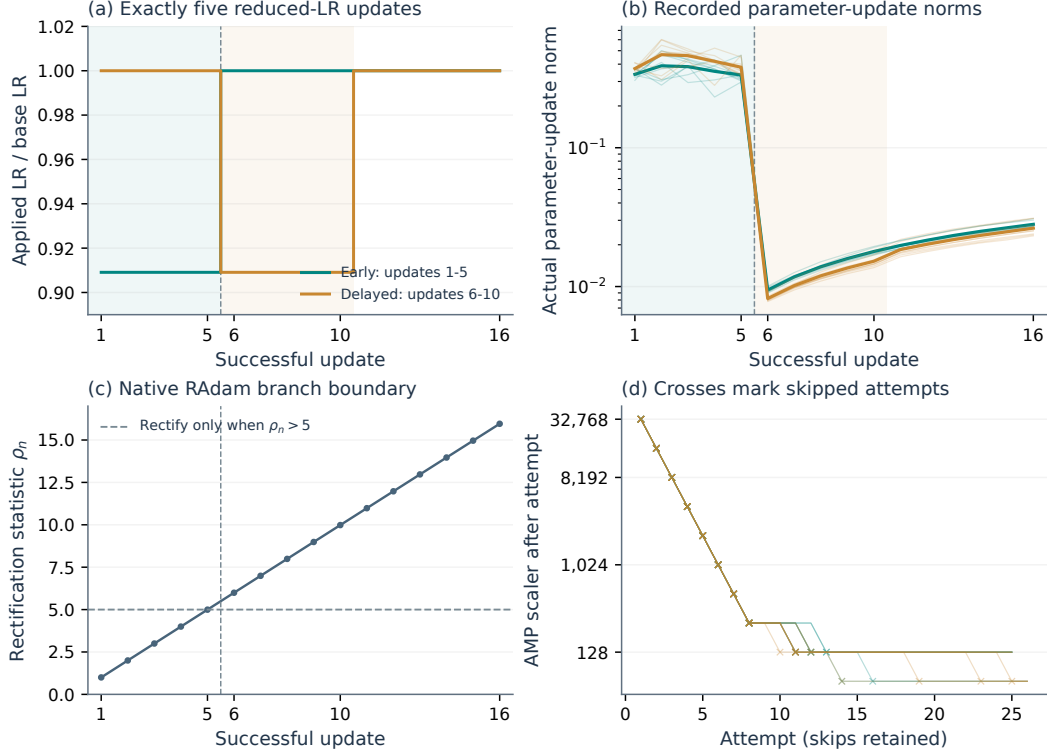


Figure 7: Executed windows and their update exposure. The top panels show the LR multipliers and recorded parameter-update norms for the first 16 successful updates; thin curves are individual paths and bold curves are arithmetic means. The lower-left panel shows RAdam’s ρ_n and its strict rectification threshold; historical early branches are reconstructed from the recorded clock and the native formula, while delayed branches are recorded directly. The lower-right panel retains attempt-level scaler values and marks AMP skips with crosses; no step occurs at those crosses. Vertical shading in the top panels identifies the two successful-update windows. Actual update magnitudes differ sharply across those windows, so equal LR multipliers and counts do not match intervention dose.

Table 12: Sum of recorded update norms in each window. All entries use the same model-parameter norm. The two windows are measured on each early and delayed path separately.

Seed	Early path		Delayed path	
	$E_{1:5}$	$E_{6:10}$	$E_{1:5}$	$E_{6:10}$
50	1.959603	0.066298	2.269626	0.054275
51	1.572802	0.070796	1.891562	0.062931
52	1.869654	0.069566	1.981548	0.068336
53	1.677424	0.074444	2.059742	0.057207
54	1.758968	0.068092	2.109287	0.060831
55	1.914078	0.065051	2.251804	0.056344
56	1.888053	0.067164	2.153815	0.056869
57	1.759548	0.069584	2.080941	0.056226

Net displacement, $\|\sum_{n \in W} (\theta_n - \theta_{n-1})\|_2$, additionally depends on update directions. The historical early logs did not retain the vectors needed to calculate it. Early net-displacement cells are explicitly missing in the archived exposure table and are never inferred from scalar norm sums. Delayed paths have recorded vector-accumulation values, but that does not fill the early-path gap. Later parameter feedback, moment histories, AMP execution, and the branch transition can all differ between placements. The observed endpoint contrast establishes the response to these two implemented operations, without selecting one of these pathways as its unique explanation.

D.3 Window provenance and execution record

The result snapshot is 3d1330f5d794d05fe63b67aa40e2dc6213f1387f; the reported scientific implementation is fcc94d26b1085011ac935084f7c1cf4c149e2947. The evaluator remains the same pinned version as in the startup pilot. The bundled, unchanged result verifier and an independent reconstruction from the 72 FID blocks reconcile all three effects, intervals, tests, and seed counts. The public manifest covers 152 result files; receipt, checkpoint-identifier, control-reuse, and telemetry bindings reconcile within the supplied records. These checks verify the public evidence package, without rehashing every private training checkpoint or rerunning GPU training.

Seeds 56/57 completed training before unrelated GPU contention prevented evaluation from starting. Their original endpoints and exception records were preserved; only empty, un-executed evaluation directories were removed before completing the original evaluations. Neither trajectory was retrained. Recorded training/evaluation process time totals 38.62 GPUh, excluding preparation, rental idle time, and the separate q128 campaign. This is an execution-accounting figure, not a benchmark of optimized training cost.

The closeout supplement now publishes the existing freeze/source-hash records and the global-budget scheduling amendment with separate raw and redacted hashes. All 351 recorded source hashes match the pinned scientific implementation. This strengthens the deployment binding; it is not a new pre-execution freeze or a retrospective training rerun. The q128-specific trajectory and duplicate-dispatch checks are detailed in Appendix G.2. The scientific source identity remains distinct from the result-publication commits.

E State interventions and continuation contrasts

Table 13: Selected state-intervention contrasts in log-FID. These rows use different complete sets and are not pooled. All displayed intervals are nominal.

Contrast	n	Mean	95% CI
M1, restarted B minus restarted A	7	-0.91661	[-2.03814, 0.20491]
M1, history $\times$ restart interaction	6	-1.00540	[-2.32150, 0.31070]
SWAP, B vs. A optimizer on A parameters	6	-0.02596	[-0.05593, 0.00401]
SWAP, B vs. A optimizer on B parameters	6	-0.01531	[-0.05428, 0.02365]
M2, between-history change in restart effect	6	1.65306	[-0.28913, 3.59525]

M1 keep-A/keep-B/restart-A/restart-B completion counts are 14/15/8/15 out of 16 each. The four-way complete set has six seeds. The swap experiments replace optimizer-source state while retaining the chosen parameter history and continuation protocol, but neither source contrast identifies a direction at the available precision. M2 compares restart at 768 versus 512 king; equal endpoint budget leaves different recovery durations. These interventions are informative about what has and has not been identified, but they do not prove a memory-decay law.

The local restart decomposition examines four retrospectively selected seeds. Clearing moments at a fixed step gives first-update norm ratios about 12.7–14.7 relative to keeping the optimizer; resetting the step and fully restarting yield different operations. Such interven-

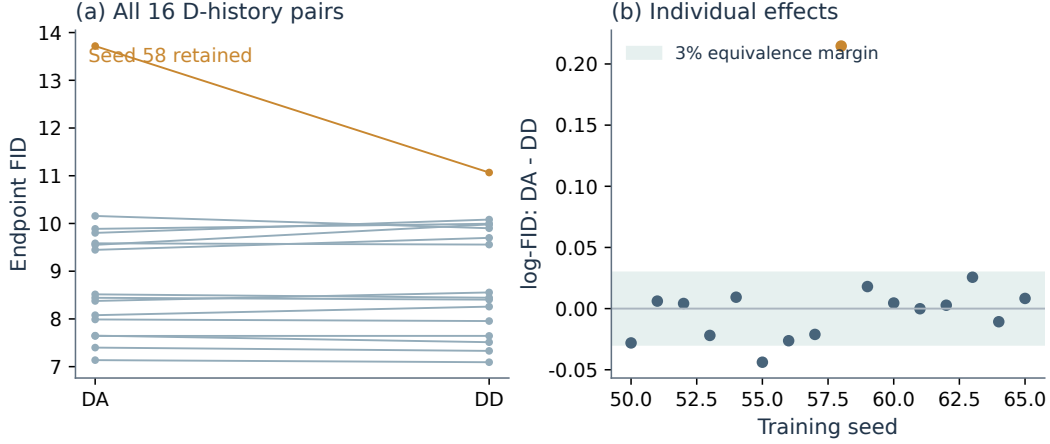


Figure 8: Restore/hold comparison from the same D prefix. All 16 paired endpoints are retained, including seed 58. The shaded band displays the prespecified 3% equivalence margin for individual log ratios; membership of individual points does not constitute a group equivalence test.

tions alter the transition map itself and cannot be treated as interventions on a single inert storage variable.

Table 14: Complete DA/DD pairs. This table includes seeds absent from the four-arm complete set.

Seed	DA FID	DD FID	DA/DD	Seed	DA FID	DD FID	DA/DD
50	9.803	10.082	0.97233	58	13.718	11.068	1.23944
51	7.137	7.093	1.00615	59	7.649	7.513	1.01818
52	7.989	7.956	1.00420	60	8.442	8.404	1.00458
53	8.078	8.257	0.97831	61	7.644	7.644	0.99988
54	7.399	7.330	1.00936	62	9.584	9.558	1.00277
55	9.552	9.980	0.95711	63	10.157	9.900	1.02597
56	9.447	9.699	0.97407	64	9.887	9.994	0.98932
57	8.376	8.555	0.97912	65	8.515	8.444	1.00835

F Longitudinal and cross-configuration evidence

F.1 Earlier four-arm learning curves

An earlier longitudinal experiment has six complete four-arm seeds, 8–13, with six checkpoints from 384 to 1,024 king and NFE1/NFE2 evaluation. Figure 9 displays arithmetic means over these same seeds. Seeds 6–7 have a different/incomplete matrix and are not silently combined with the six complete trajectories.

The descriptive NFE2 normalized raw-FID areas are 32.740, 18.852, 31.942, and 21.221 for A/B/C/D. D–A is favorable in 6/6 seeds and B–A in 5/6; the corresponding endpoint means are 3.045, 2.914, 3.082, and 2.930. A normalized curve area here means the trapezoidal integral over 384–1,024 king divided by the interval length. It was not the prespecified primary endpoint, and it is not numerically interchangeable with a log-FID curve area used in other studies. The NFE1/NFE2 curves provide motivation for studying trajectories without establishing a new general budget law.

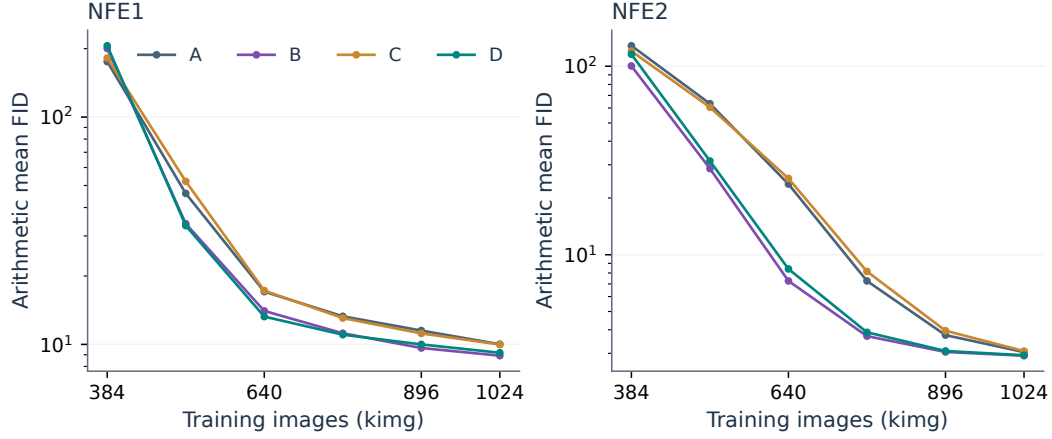


Figure 9: Earlier four-arm trajectories, complete seeds 8–13. Error magnitudes early in tuning make the log axis useful. These experiments keep their assigned objective throughout; their estimand differs from the later common-continuation study.

Table 15: Additional scope studies. Their outcomes do not share a single estimand or validity status.

Design	Observed result	Interpretation
Fresh q_{256} crossed switch	11 complete pairs; mean history contrast -0.07557 , CI $[-0.15532, 0.00419]$	Direction unresolved under its frozen analysis
Short pulse and chase	Prespecified primary contrast satisfies its 3% equivalence criterion	Informative null for that exposure and readout
Fresh q_{128} history	8 analyzed seeds; mean -0.03202 , CI $[-0.08577, 0.02173]$	Direction unresolved; seed replacement and AMP-gate deviations prevent confirmatory interpretation
Switch-time scan	12 seeds; prespecified Page ordering test $p = 0.6719$	No established ordering or universal critical window

F.2 Complete ImageNet readouts

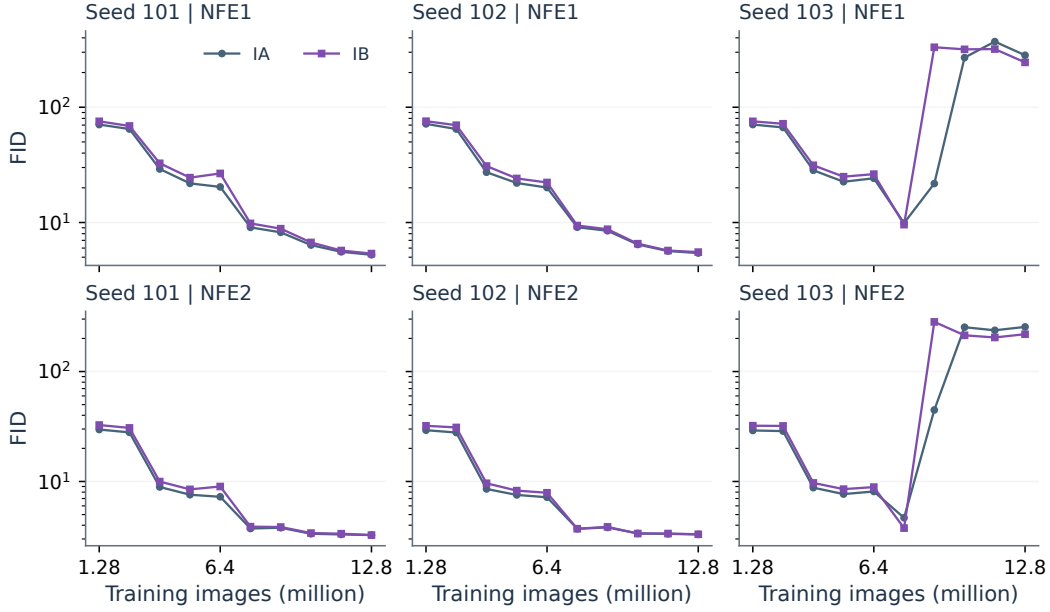


Figure 10: A configuration boundary for enlarged-gap gains. Three paired ImageNet- 64×64 seeds, ten checkpoints, and two NFEs are retained, including deterioration in both seed-103 paths. IA uses gap factor 1 and IB uses 1.1. The log scale accommodates the quality range. This configuration uses $c = 0.06$ and gap-independent snrpk timestep weighting, and differs from the main experiment in data, architecture, and other settings. Repeated readouts do not increase the independent training-seed count.

Tables 16–17 retain every recorded paired checkpoint for seeds 101–103. Ten checkpoints, two arms, and two NFEs give 120 evaluation cells; the independent training replication count is three pairs. The favorable early IA comparisons are repeated observations within those pairs. Late deterioration is retained, and no best-checkpoint selection is used as a primary result.

Table 16: All ImageNet FID readouts; IA/IB denote gap factors 1/1.1.

Seed	king	IA NFE1	IB NFE1	IA NFE2	IB NFE2
101	1280	70.730	75.483	29.679	32.613
101	2560	64.728	68.780	27.959	30.697
101	3840	29.127	32.643	8.912	9.981
101	5120	21.858	24.503	7.578	8.477
101	6400	20.357	26.653	7.256	8.998
101	7680	9.071	9.823	3.734	3.883
101	8960	8.219	8.841	3.795	3.852
101	10240	6.373	6.716	3.345	3.394
101	11520	5.567	5.714	3.298	3.349
101	12800	5.249	5.375	3.255	3.255
102	1280	71.747	75.673	29.264	32.093
102	2560	64.704	69.877	27.874	31.013
102	3840	27.300	30.896	8.547	9.627
102	5120	22.012	24.119	7.546	8.256
102	6400	20.098	22.246	7.186	7.895
102	7680	9.101	9.415	3.717	3.715
102	8960	8.481	8.740	3.819	3.862
102	10240	6.447	6.549	3.368	3.361
102	11520	5.652	5.711	3.338	3.361

Seed	king	IA NFE1	IB NFE1	IA NFE2	IB NFE2
102	12800	5.432	5.537	3.301	3.309
103	1280	70.773	75.442	29.108	32.158
103	2560	66.805	71.889	28.728	32.009
103	3840	28.389	31.367	8.784	9.703
103	5120	22.576	24.981	7.679	8.516
103	6400	24.188	26.283	8.108	8.893
103	7680	9.884	9.584	4.674	3.767
103	8960	21.773	331.368	44.715	282.814
103	10240	269.538	317.867	253.039	213.578
103	11520	371.026	319.572	237.166	204.166
103	12800	282.502	244.955	254.736	218.597

Table 17: All corresponding ImageNet KID readouts. These use the same features as FID and are not independent replications.

Seed	king	IA NFE1	IB NFE1	IA NFE2	IB NFE2
101	1280	0.05126	0.05536	0.01806	0.02005
101	2560	0.04603	0.04922	0.01672	0.01842
101	3840	0.01873	0.02124	0.00489	0.00558
101	5120	0.01371	0.01554	0.00409	0.00462
101	6400	0.01269	0.01699	0.00394	0.00483
101	7680	0.00581	0.00620	0.00216	0.00223
101	8960	0.00521	0.00553	0.00224	0.00219
101	10240	0.00389	0.00414	0.00189	0.00197
101	11520	0.00335	0.00346	0.00198	0.00202
101	12800	0.00313	0.00320	0.00196	0.00189
102	1280	0.05168	0.05490	0.01743	0.01925
102	2560	0.04658	0.05073	0.01681	0.01887
102	3840	0.01755	0.02004	0.00479	0.00549
102	5120	0.01380	0.01518	0.00414	0.00452
102	6400	0.01250	0.01384	0.00386	0.00419
102	7680	0.00543	0.00563	0.00204	0.00198
102	8960	0.00545	0.00553	0.00234	0.00231
102	10240	0.00402	0.00404	0.00204	0.00199
102	11520	0.00355	0.00353	0.00209	0.00210
102	12800	0.00336	0.00338	0.00206	0.00200
103	1280	0.05097	0.05491	0.01763	0.01949
103	2560	0.04846	0.05268	0.01742	0.01948
103	3840	0.01802	0.01996	0.00477	0.00531
103	5120	0.01423	0.01586	0.00423	0.00477
103	6400	0.01525	0.01662	0.00434	0.00479
103	7680	0.00619	0.00595	0.00252	0.00212
103	8960	0.01732	0.38472	0.04286	0.28751
103	10240	0.26849	0.35220	0.22787	0.22443
103	11520	0.45382	0.35964	0.27935	0.24254
103	12800	0.30823	0.26088	0.29864	0.26976

G Complete fresh q128 results and bounded diagnostics

The cohort comprises all seeds 301–308 and all four fresh trajectories per seed, with no result-driven replacements. Table 18 reports every endpoint and contrast. For each seed and arm, $\bar{Y}$ is the mean of three log-FID50k values. The primary test is the paired-t test of $S = (C - M)/2$; M/C use Holm across those two tests, and H is supportive. Exhaustive sign flips for S/M/C are sensitivity analyses under sign symmetry, not design-based randomization tests. Neither blocks nor metrics increase the number of independent training seeds.

Table 18: All fresh q128 endpoints and seed-level log contrasts. Arm columns contain $\exp(Y)$, the geometric mean of three FID50k blocks; contrast columns contain log-FID differences. These values are recomputed from the unchanged 96-block table.

Seed	AA	DA	$A_{\downarrow 5}$	$D_{\uparrow 5}$	M	C	H	S
301	9.1475	8.3261	8.4747	9.2479	-0.07639	+0.10499	-0.09408	+0.09069
302	8.4061	7.8880	8.2529	8.1892	-0.01840	+0.03747	-0.06361	+0.02793
303	7.9640	8.6829	8.4771	8.0728	+0.06244	-0.07285	+0.08642	-0.06764
304	9.4385	9.4047	9.2519	8.9091	-0.01996	-0.05414	-0.00358	-0.01709
305	7.7838	7.6307	8.0216	8.1578	+0.03009	+0.06679	-0.01987	+0.01835
306	8.4062	8.6694	8.6185	8.5044	+0.02494	-0.01921	+0.03083	-0.02207
307	9.4320	9.4180	9.6343	9.4632	+0.02123	+0.00479	-0.00148	-0.00822
308	9.2294	9.1750	9.3522	9.5216	+0.01322	+0.03708	-0.00592	+0.01193

G.1 Local update response and its limits

For a successful-update window W , the scalar path-length reading is $\sum_{k \in W} \|\Delta\theta_k\|_2$ and the net displacement is $\|\sum_{k \in W} \Delta\theta_k\|_2$. The latter comes from the existing float64 vector accumulator, not from scalar norms. The original vectors are not available in this small-file supplement, so its net-displacement readings are preserved recorded statistics rather than independently reconstructed vectors. Windows 1–5 and 6–10 were already recorded; no additional window was selected using endpoint quality.

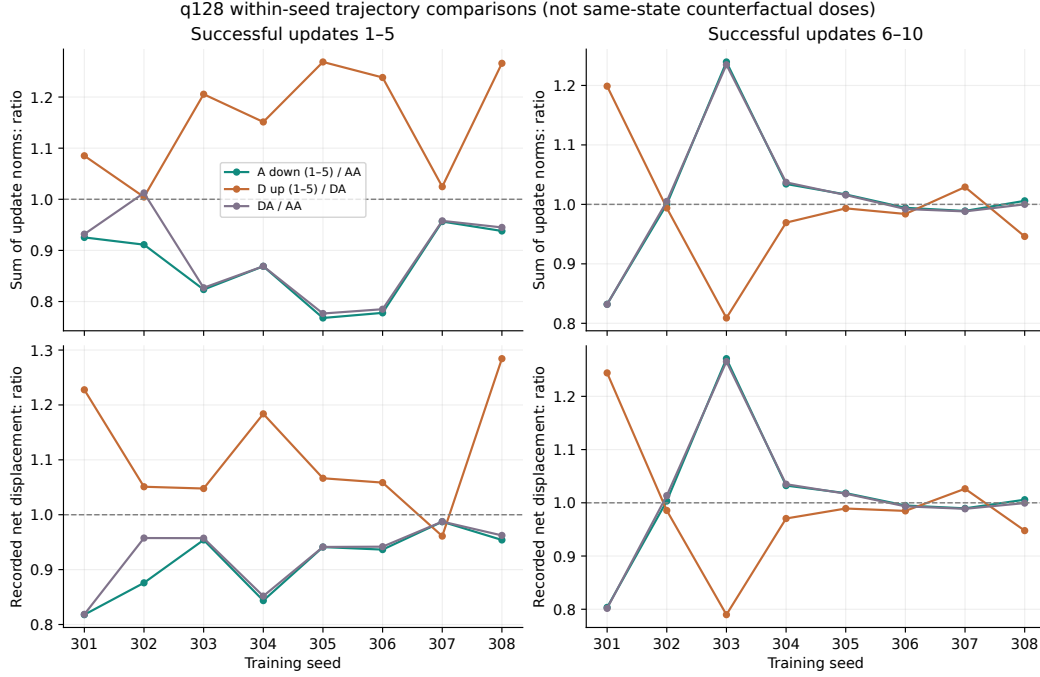


Figure 11: Local response in every fresh q128 seed. Ratios compare each intervened path with its stated reference within the same training seed. Top: sums of actual update norms. Bottom: existing net-displacement readings. Columns retain the predefined adjacent successful-update windows. These paths can differ in parameters, moments and AMP events; the ratios are not same-state counterfactual doses. Lines connect seed labels for readability and do not describe a time series or fitted relationship.

The first-window scalar sums decrease for all eight A-reduction paths and increase for all eight D-compensation paths. Net displacement has one different direction on the D side,

illustrating why norm sums cannot replace vector accumulation. Later-window ratios are heterogeneous. No seed selection, quality-prediction model, threshold search or update-FID correlation search is used. The existing material does not explain why this local response lacks a stable endpoint benefit.

The q256 early/delayed readings remain in Table 12; the supplement places them alongside all q128 readings as descriptive references. All 16 missing q256 early net-displacement cells remain missing. Since q and cohort change together, neither differences in movement nor endpoint effects identify a causal q interaction.

Table 19: Complete q128 movement readings, all eight seeds and four arms, for the two existing windows. “Sum” is the sum of update norms; “Net” is the recorded vector-sum norm.

Seed	Arm	Sum 1–5	Net 1–5	Sum 6–10	Net 6–10
301	AA	0.956371	0.638786	0.075649	0.074086
301	DA	0.891166	0.522870	0.062929	0.059393
301	$A_{\downarrow 5}$	0.885036	0.522471	0.062943	0.059527
301	$D_{\uparrow 5}$	0.967023	0.641896	0.075437	0.073887
302	AA	1.032667	0.588143	0.063008	0.059212
302	DA	1.045479	0.563167	0.063321	0.060008
302	$A_{\downarrow 5}$	0.941008	0.515174	0.063010	0.059410
302	$D_{\uparrow 5}$	1.050152	0.591873	0.062939	0.059144
303	AA	0.844699	0.556199	0.062233	0.058674
303	DA	0.698486	0.532440	0.076847	0.074241
303	$A_{\downarrow 5}$	0.695549	0.530647	0.077137	0.074564
303	$D_{\uparrow 5}$	0.842038	0.557862	0.062167	0.058637
304	AA	1.087490	0.879785	0.076896	0.076166
304	DA	0.945079	0.749368	0.079746	0.078828
304	$A_{\downarrow 5}$	0.944939	0.742149	0.079532	0.078630
304	$D_{\uparrow 5}$	1.088023	0.887089	0.077298	0.076503
305	AA	0.768483	0.509177	0.061884	0.058572
305	DA	0.596780	0.479290	0.062856	0.059574
305	$A_{\downarrow 5}$	0.590064	0.479173	0.062917	0.059639
305	$D_{\uparrow 5}$	0.757090	0.511133	0.062434	0.058935
306	AA	0.704334	0.483697	0.065098	0.061477
306	DA	0.552870	0.455544	0.064589	0.061067
306	$A_{\downarrow 5}$	0.547851	0.452972	0.064722	0.061168
306	$D_{\uparrow 5}$	0.684591	0.482199	0.063548	0.060138
307	AA	0.976255	0.770380	0.079128	0.078127
307	DA	0.935073	0.760734	0.078192	0.077245
307	$A_{\downarrow 5}$	0.933747	0.760546	0.078255	0.077313
307	$D_{\uparrow 5}$	0.958117	0.731099	0.080469	0.079286
308	AA	1.040063	0.741739	0.074520	0.073428
308	DA	0.982823	0.713781	0.074525	0.073389
308	$A_{\downarrow 5}$	0.975592	0.707651	0.074965	0.073855
308	$D_{\uparrow 5}$	1.244252	0.916725	0.070507	0.069553

G.2 Scientific identity and evidence coverage

The reviewed PR113 result commit is 42871769be317ff6baa1bd6e030bec120c22d9371; the scientific implementation is fcc94d26b1085011ac935084f7c1cf4c149e2947, and the evaluator is d6aba02fb88e9db0993623895eb2228ed717d810. The source Overleaf edition was already bound to PR112 3d1330f5d794d05fe63b67aa40e2dc6213f1387f. Publication commits are not substituted for run-code identities.

The existing public verifiers pass unchanged. The additional audit binds each canonical initial receipt to all four same-seed initial model, EMA, optimizer, scaler, RNG and sampler hashes; checks the effective full scientific configuration, actual manifests and endpoint seals; and retains continuous telemetry through successful update 16, including all intervening skips. All 32 paths pass the checked startup conditions. The 763 attempts include 251 skips and 512 successful updates. Successful calls report checks of all active parameter clocks;

skip rows carry the unchanged successful-call counter and are not mislabeled as fresh direct parameter-clock measurements. LR multipliers on skip rows denote no applied optimizer update.

All 32 readable process receipts are byte-identical, by original SHA256, to the archived outcome identities; this provides a readable equivalent where the outcome filename itself is permission-restricted. The frozen worker checks the endpoint’s actual loss phase/q, optimizer state, and single own- E_{512} initialization before issuing PASS. The small-file seals bind q128 at 512 king, branch initialization and 1024 king. An existing explicit before/after full-state boundary check is available for seed302 DA. The other 31 boundaries were not independently rehashed from checkpoint tensors in this zero-GPU closeout; this narrower evidence coverage is recorded as unverified, not as an execution failure. Existing engineering checks for seeds 99001/99002 are reused, with no rerun.

G.3 Three excluded partial duplicate launches

The retained routing amendment assigns the affected formal jobs to their canonical hosts before those jobs begin training or produce quality results. Later host-permit handling inadvertently launches three partial duplicates. They are stopped, separately archived, and do not replace the formal trajectories. Their recorded process costs sum to 0.400088 GPUh and are included in the original campaign ledger. Two canonical jobs had already produced some quality results before the duplicate launches; the exclusion evidence rests on the earlier host assignment, not on a claim that incident handling preceded every outcome. No duplicate evaluation appears in the retained duplicate inventories.

The supplement publishes the routing, incident, stopping, archive and cost records using stable host aliases, with raw-file hashes distinguished from public-file hashes and explicit extraction boundaries. These materials are an after-the-fact evidence supplement, not a newly claimed pre-execution record. No training, sample generation or FID/KID generation was performed for the closeout: additional training/generation GPU cost is zero. The work stops at the limits of the existing records.